

CHAPTER 1

LEARNING OUTCOME 1:

**CLEAN AND MAINTAIN
KITCHEN TOOLS, AND
PREMISES**

Objectives

At the end of the lesson, you are expected to:

- a) Clean, sanitize and store kitchen tools and equipment;
- b) Clean and sanitize kitchen premises.

Cleanliness is vital in every kitchen where food is prepared, cooked and served. In order to avoid food contamination, kitchen tools, equipment and other utensils used in the preparation of food as well as its premises should be cleaned and sanitized, and store properly after each use.

Definition of Terms

Bleach

Sodium hypochlorite used in diluted bleach solution in water is used to sanitize and clean kitchen areas and surfaces.

Cleaning Cloth

Cloth that is used for cleaning. Not to be cleaning cloth used for food production.

Cleaning Equipment

Any piece of equipment designed for the specific use of cleaning including vacuum cleaners, mops, etc.

Cleaning Schedule

A plan to schedule cleaning of all equipment and utensils within the premises used to produce food. Also states when to clean the building and surround where the food production carried out.

Cleaning Utensils

Small pieces of equipment like brooms, scourers, scrubbing brushes, mops, buckets.

Detergent

Specific liquid, sprays, powders, etc. used in the purpose of cleaning.

Kitchen Equipment

Large equipment used to cook food like stoves, cool rooms, normally fixed in place.

Kitchen Utensils

Small utensils used in the kitchen for food production, spoons, knives, ladles.

Legislation

Pieces of law or regulations.

Production Area

Where food is processed for human consumption.

Personal Protection Equipment (PPE)

Gloves, face mask, eye protection, aprons, hats: clothing or apparel that protect person from harm while using harsh chemicals.

Sanitize

To make clean, remove bacteria to safe level.

Storage Area

Where food is stored prior to processing and after processing.

Waste Disposal

Removal of rubbish from food production area.



LEARNING OUTCOME 2:

**CLEAN AND SANITIZE
KITCHEN TOOLS AND
EQUIPMENT**

Expected Learning Outcomes: At the end of this lesson, you are expected to:

- LO 1. Recognize kitchen tools and equipment to be cleaned and sanitized;
- LO 2. Identify the chemicals to be utilized in cleaning and sanitizing kitchen tools and equipment;
- LO 3. Prepare cleaning agents in accordance with the manufacture's Instruction/procedure;
- LO 4. Clean and sanitize kitchen tools and equipment; and
- LO 5. Store cleaned kitchen tools and equipment.

Cleaning and sanitizing kitchen tools and equipment must be part of the standard operating procedures that make up your food safety program. Improper cleaning and sanitizing kitchen surfaces allow harmful microorganisms to be transferred from one another



Uses of kitchen tools and equipment's

The following are list of cooking materials, kitchen utensils and equipment that are commonly found in kitchen

Cooking Materials

Aluminum is mostly used in the kitchen and the most popular because it is lightweight, attractive and less expensive.



Stainless Steel is the most popular material used for tools and equipment, but it is more expensive. It is easier to clean and shine and will not wear out easily.



Glass is used for salad making and dessert but not practical for top or surface cooking. Great care is needed to ensure for long shelf life.



Can opener is used to open food containers.



Colanders also called a vegetable strainer, are essential for various tasks from cleaning vegetables to straining pasta or contents.



Cast Iron is durable but must be kept oiled to avoid rusting. Salad oil with no salt or shortening can be rubbed inside and out and dried. Wash with soap (not detergent) before using.



Double boiler is used when temperature must be kept below boiling, such as for egg sauces, puddings; it is used to keep food warm without overcooking.



Teflon is a special coating applied inside aluminum or steel pots and pans. It prevents food from sticking to the pan.



Kitchen Tools

Plastic and hard rubber knives are used for cutting and chopping. They are duller than knives. Plastics are greatly durable and cheap but may not last long.



Funnels are used to fill jars, made of various sizes of stainless steel, aluminum, or of plastic.



Garlic Press is a kitchen tool which is specifically designed for the purpose of pulping garlic.



Graters are used to grate, shred, slice and separate vegetables such as carrots, cabbage and cheese.



Kitchen shears they are practical for opening food packages, cutting tapes or strings or simply remove labels or tags from items.



Potato masher is used for mashing cooked potatoes, turnips, carrots or other soft cooked vegetables.



Rotary egg beater is used for beating small amount of eggs or batter. The beater should be made of stainless steel.



Scraper is a rubber or silicone tool used to blend or scrape the food from the bowl.



Serving spoons are small, shallow bowl on a handle used in preparing, serving, or eating food.



Serving tongs used to grab or transfer food items.



Spatula is used to level off ingredients when measuring and to spread frostings and sandwich fillings.



Spoons are solid, slotted, or perforated Which are made of stainless steel or plastic. The solid ones are used to spoon liquids over foods and to lift foods, including the liquid out of the pot.



Temperature scales are used to measure heat intensity. Different thermometers are used for different purposes in food preparation – for meat, candy or deep-fat frying and other small thermometers are hanged or stand in ovens or refrigerators to check the accuracy of the equipment's thermostat.



Wire whisk is used for eggs or batter, and for blending gravies, sauces, and soups.



Wooden spoons are made of hard wood which are used for creaming, stirring, and mixing.



Household Scales are used to weigh large quantity of ingredients in kilos, commonly in rice, flour, sugar, legumes or vegetables and meat up to 25 pound.



Kitchen Equipment

Refrigerator or freezer is an insulated box, equipped with refrigeration unit and a control to maintain the proper inside temperature for food storage.



Chiller used for displaying food items.



Oven is a chamber or compartment used for cooking, baking, heating, or drying.



Microwave ovens are used for cooking or heating food.



Blenders are used to chop, blend, mix, whip, puree, grate, and liquefy all kinds of food. A blender is a very useful appliance. They vary in the amount of power. (voltage/wattage)



Measuring Tools

Measuring cup for liquid ingredients is commonly made up of heat-proof glass and transparent so that the liquid can be seen



Chemicals used in cleaning and sanitizing kitchen tools and equipment

Cleaning is the process of removing food and other types of soil from a surface, such as a dish, glass, or cutting board. Cleaning is done with a cleaning agent that removes food, soil, or other substances. The right cleaning agent must be selected because not all cleaning agents can be used on food-contact surfaces. (A food-contact surface is the surface of equipment or utensil that food normally comes into contact.) For example, glass cleaners, some metal cleaners, and most bathroom cleaners cannot be used because they might leave an unsafe residue on the food contact surface. The label should indicate if the product can be used on a food contact surface. The right cleaning agent must also be selected to make cleaning easy

Cleaning Compound

Detergents are cleaning agents, solvents or any substance used to wash tablewares, surfaces, and equipment's. (Example: soap, soap powders, cleaners, acids, volatile solvents and abrasives.)

Solvent Cleaners commonly referred to as degreasers, they are used on surfaces where grease has burned on. Ovens and grills are examples of areas that need frequent degreasing. These products are alkaline based and are formulated to dissolve grease.

Acid Cleaners used periodically in removing mineral deposits and other soils that detergents cannot eliminate such as scale in washing machines and steam tables, lime buildup on dishwashing machines and rust on shelving. (Example: phosphoric acid, nitric acid, etc.) These products vary depending on the specific purpose of the product.

Abrasives are generally used to remove heavy accumulations of soil that are difficult to remove with detergents, solvents and acids. These products must be carefully used to avoid damage to the surface being cleaned.

Other chemicals used for cleaning and/or sanitizing kitchen equipment and utensils are the following:

1. Ammonia
2. Dish washing liquid
3. Chlorine
4. Carbonic Acid
5. Timsen
6. Disinfectants and Soaps

Steps in Dishwashing

1. **Wear rubber gloves** if you have dry hands or skin problem if you are wearing long sleeves, roll them up or put them under gloves. Wear aprons too.



2. **Scrape** all the large pieces of food on the dishes and place it in a compost bin or garbage can.



3. **Stack** the dishes in the proper order namely: glassware, silverware, chinaware, and utensils. Stack them to the right of the sink so that work progresses from right to left.



4. **Fill** the sink with water and add a considerable amount of detergent. The hotter the water, the better it's sanitizing and grease-cutting properties but use tolerable heat (66°C /150°F or above) so not to scald yourself. Use rubber gloves.



5. **Wash** the lightest soiled items first. Start with glasses, cups, and flatware. Soap each piece individually and rinse in hot water.



6. **Wash plates, bowls, and serving dishes.** Remember to scrape these items before washing.



7. **Wash pots and pans last.** Soak them first. Wash the pans thoroughly and don't forget to clean the bottoms. If anything was burnt or overcooked to pots or casserole dishes, put a little extra soap and water in it and let it stand while you wash the other dishes. Take note that any oil residue left will lead to burnt food during the next cooking session.

8. **Lay** your dishes out on a rack to air-dry or wipe them clean with a towel.



9. **Run a hand over the dish** to ensure that they are thoroughly cleaned. There should be no visible matter and no "greasy" feel. If there are still some grease remaining, consider rewashing the item.

10. **Rinse out brush**, sponge and allow to dry. Sterilize your equipment often using boiling water with bleach. When a sponge or brush starts to smell unpleasant, throw it away.

11. **Wipe down the sink and your tools**, Wipe down the sink, dish drainer, and dishpan. Any rags, dish cloths or sponges need to be left out to air dry, or thrown into the washing machine. Remember to replace sponge and rags frequently.



Steps in Dishwashing using Dishwashing Machine

Fill your dishwasher logically.



Things to consider: Many dishwashers have a variety of spacings to accommodate large bowls, small bowls, utensils, dishes, and plates of varying sizes—on the bottom shelf. The top shelf is given to shorter items such as glasses, mugs, jars, and long utensils such as spatulas and stirring spoons.

Don't crowd the dishes. Fill your dishwasher full, but not crammed—you'll maximize the cleaning capacity of your dishwasher and minimize excessive water usage.



Add detergent. Fill the detergent dispenser with cleanser either liquid or powder and close it up.

- -You can add extra cleanser in a secondary cup should your dishes be excessively filthy.
- Add a rinsing agent to prevent spotting, if necessary.



Turn it on. Set the timer as necessary. A shorter time for lightly-soiled dishes, or a longer time for heavily-soiled pots, pans, and dishes.

Dry the dishes. You can use a heated dry (but be careful of plastic dishes or containers), or use air



How to Clean, Remove Stains, Sanitize and Store

The kitchen cutting board gets a lot of use and this means that it gets a lot of exposure to bacteria. Proper cleaning of the cutting board is essential to your good health. Whether you use a wood or plastic cutting board, you should clean and sanitize it after use.

Clean the Cutting Board

After you used the cutting board for slicing, dicing, or chopping all kinds of neat goodies, use a metal scraper or spatula to scrape away any remaining bits and pieces of food. Throw the scraps into the garbage disposal, garbage receptacle, or trash bin.

Scrub the board with hot, soapy water thoroughly. If your dishwasher reaches a temperature of at least 165°F, then you can probably place a high density plastic cutting board into the

dishwasher. Moreover, if your dishwasher has an antibacterial cycle, use it to wash the cutting board. Otherwise, scrub it by hand. Allow the board to air dry. Removing Stains from the Cutting Board

Removing Stains from the Cutting Board

To remove stains from the cutting board, you can use the following procedure: wet the stained area with water and sprinkle it with salt and allow the salt to melt undisturbed for twenty-four hours.

Rinse the salt from the cutting board with clean water. Using the salt and clean water, create a paste. Use a clean nylon scrubbing sponge or a clean toothbrush to scour or scrub the paste on the stained area of the cutting board. Rinse the area clean with fresh water. Repeat the procedure to guarantee that you have removed all of the stain. Rinse the board clean. Scrub the cutting board with hot, soapy water and rinse with clean water. Allow it to air dry.

Sanitizing the Cutting Board

Plastic and wooden cutting boards can be sanitized using a diluted liquid chlorine bleach solution. For this solution, combine one teaspoon of bleach to one quart of water. Pour the solution onto the entire surface area of the board and allow it to sit undisturbed for several minutes. Rinse the board clean with water. Allow it to air dry or use a clean cloth to dry it.

If you prefer, you may use a vinegar solution in place of the bleach solution. Simply combine one part vinegar to five parts water. Use this solution in the same manner as the one explained for the bleach solution.

Storing the Cutting Board

Once the cutting board has completely dried, store it vertically or in an upright position. This helps to avoid moisture from getting trapped underneath the board and the accumulation of dust or grime.

Fundamental Cleaning Procedures

1. *Scrape and Pre-rinse* – soiled equipment surfaces are scraped and rinsed with warm water to remove loose food soils.

2. *Cleaning Cycle* – the removal of residual food soils from equipment surfaces is based on the manipulation of the four basic cleaning factors and the method of cleaning. Typically, alkaline chemical solutions are used for the cleaning cycle.

3. *Rinse* – rinse all surfaces with cold to hot water, depending on the temperature of the cleaning cycle, to thoroughly remove all remaining chemical solution and food soil residues.

4. *Acid Rinse* – a mild acid rinse of the equipment neutralizes any alkaline residues left and removes any mineral soil present.

5. *Sanitize* – all equipment surfaces are rinsed or flooded with a sanitizing agent.

Both time and chemical concentration are critical for optimum results

FACTORS THAT INFLUENCE THE CLEANING PROCESS

Soil – varying degrees of food soil will be deposited on the equipment during production. These food soils will require complete removal during the cleaning process and will affect the cleaning compound used, along with the method of cleaning.

Time – the longer a cleaning solution remains in contact with the equipment surface, the greater the amount of food soil that is removed. More time in contact with the soil reduces the chemical concentration requirements.

Chemical concentrations – it varies depending on the chemical itself, type of food soil, and the equipment to be cleaned. Concentration will normally be reduced as time and temperature are increased.

Temperature – soils are affected by temperature in varying degrees. In the presence of a cleaning solution most soils become more readily soluble as the temperature increases.

Mechanical force – is as simple as hand scrubbing with a brush or as complex as turbulent flow and pressure inside a pipeline. This aids in soil removal and typically reduces time, temperature and concentration requirements.

Water – minerals in hard water can reduce the effectiveness of some detergents or sanitizers. Water pH ranges generally from pH5-8.5. However, highly acidic water may require additional buffering agents. Water used for cleaning and sanitizing must be potable and pathogen free.

Finally you've done the cleaning but ooops... not so fast because after cleaning the kitchen tools and equipment the next thing you will do is to sanitize them. How? The following topics are the methods of sanitizing kitchen tools and equipment after cleaning them. Sanitizing is done using heat, radiation, or chemicals. Heat and chemicals are commonly used as a method for sanitizing in a restaurant than radiation. The item to be sanitized must first be washed properly before it can be properly sanitized. Some chemical sanitizers, such as chlorine and iodine, react with food and soil and so will be less effective on a surface that has not been properly cleaned.

Methods of Sanitizing

Thermal Sanitizing

It involves the use of hot water or steam. There are three methods of using heat to sanitize surfaces – steam, hot water, and hot air. Hot water is the most common method used in restaurants. If hot water is used in the third compartment of a three-compartment sink, it must be at least 171°F (77°C). If a high-temperature ware washing machine is used to sanitize cleaned dishes, the final sanitizing rinse must be at least 180°F (82°C). For stationary rack, single temperature machines, it must be at least 165°F (74°C). Cleaned items must be exposed to these temperatures for at least 30 seconds.

Chemicals

Approved chemical sanitizers are chlorine, iodine, and quaternary ammonium. Different factors influence the effectiveness of chemical sanitizers. The three factors that must be considered are:

- a. Concentration. The presence of too little sanitizer will result in an inadequate reduction of harmful microorganisms. Too much can be toxic.
- b. Temperature. Generally, chemical sanitizers work best in water that is between 55°F (13°C) and 120°F (49°C).
- c. Contact time. In order for the sanitizer to kill harmful microorganisms, the cleaned item must be in contact with the sanitizer (either heat or approved chemical) for the recommended length of time

Characteristics of ideal chemical sanitizer

- Approved for food contact surface application
- Have a wide range or scope of activity
- Destroys microorganisms rapidly
- Stable under all types of conditions
- Tolerate a broad range of environmental conditions
- Readily solubilizes and possesses some detergency
- Low in toxicity and corrosively
- Inexpensive
- Heat sanitizing has several advantages over chemical sanitizing agents?
- can penetrate small cracks and crevices
- non-corrosive to metal surfaces
- non-selective to microbial groups
- leaves no residues
- easily measurable

Advantages and Disadvantages of Different Chemical Sanitizers

Chemical	Concentration	Contact Time	Advantage	Disadvantage
Chlorine	50 ppm in water between 75 and 100°F	7 seconds	Effective on a wide variety of bacteria; highly effective; not affected by hard water; generally inexpensive	Corrosive, irritating to the skin, effectiveness decreases with increasing pH of solution; deteriorates during storage and when exposed to light; dissipates rapidly; loses activity in the presence of organic matter
Iodine	12.5-25 ppm in water that is at least 75°F	30 seconds	Forms brown color that indicates strength; not affected by hard water; less irritating to the skin than is chlorine; and activity not lost rapidly in the presence of organic matter.	Effectiveness decreases greatly with an increase in pH (most active at pH 3.0; very low acting at pH 7.0); should not be used in water that is at 120°F or hotter; and might discolor equipment and surfaces.
Quaternary Ammonium Compounds	Up to 200 ppm in water that is at least 75°F	30 seconds	Nontoxic, odorless, colorless, noncorrosive, nonirritating; stable to heat and relatively stable in the presence of organic matter; active over a wide pH range	Slow destruction of some microorganisms; not compatible with some detergents and hard water

“Of course you will not just leave the cleaned and sanitized kitchen tools and equipment in a place that is exposed to dust and/or microbes, right? The next topic will give you the complete idea on proper storage of kitchen tools and equipment”

Proper Storage of Kitchen Tools and Equipment

Proper storage and handling of cleaned and sanitized equipment and utensils are very important to prevent recontamination prior to use. Cleaned and sanitized equipment and utensils must be:

- o Stored in clean storage areas; and
- o Handled properly to minimize contamination of food contact surface:

9 Steps in Organizing Kitchen Cabinets

Pretend it has a glass door and that everyone is going to see what's inside.

1. Remove all the equipment and scrub shelves with soapy water.
2. Think about what you reach most often and make sure it gets a position that is easy to reach.
3. Take a cabinet full of glasses and line them up by color. Make sure all of the front are facing out and straight. (Jeff Lewis-Style)

4. Take a step back after one shelf is done and make someone else look at what you have done.
5. They should be stored in a clean dry place adequately protected against vermin and other sources of contamination.
6. Cups, bowls, and glasses must be inverted for storage.
7. When not stored in closed cupboards or lockers, utensils and containers must be covered or inverted whenever possible. Utensils must be stored on the bottom shelves of open cabinets below the working top level.

8. Racks, trays and shelves must be made of materials that are imperious, corrosive- resistant, non-toxic, smooth, durable and resistant to chipping.

9. Drawers must be made of the same materials and kept clean. Full-lined drawers are not acceptable, but the use of clean and removable towels for lining drawers is acceptable.



LEARNING OUTCOME 3:

**CLEAN AND SANITIZE
KITCHEN PREMISES**

Objectives:

The lesson deals with the various types of chemicals and equipment for cleaning and sanitizing premises and first aid procedure for accidents caused by chemicals. Specifically after this module you will be able to:

- 1.3.1 Recognize kitchen premises to be cleaned and sanitized;
- 1.3.2 Classify and describe the uses of cleaning agents;
- 1.3.3 Clean the kitchen area in accordance with food safety and occupational health regulations;
- 1.3.4 Use cleaning agents in sanitizing kitchen premises; E. Follow safety and first aid procedures.

Cleaning your kitchen's working premises regularly is important to keep it look its best and make it free from germs and bacteria that usually accumulate in the kitchen area during food preparations. Several surfaces around the kitchen such as walls, floors, shelves and other surfaces must always be cleaned and sanitized safely using the proper materials to reduce environmental deficiencies



Types of Sanitizers and Disinfectants

There are various types of chemicals use for sanitizing and disinfecting equipment, and first aid procedures for accidents caused by chemicals.

Chemicals

- A. chlorine
- B. carbolic acid
- C. ammonia
- D. detergents
- E. dishwashing liquid
- F. timsen
- G. soap
- H. alcohol
- I. boric acid

Heat Sanitizer

- A. Hot water
- B. Steam
- C. Dry heat
- D. UV light (ultraviolet light)
- E. Filtration

Procedure for disinfecting

- A. Preliminary cleaning is required.
- B. Apply solution to hard, non-porous surface thoroughly wetting it with cloth, mop, and sponge.
Treated surface must remain wet for 10 minutes. Wipe with dry cloth.
- C. Sponge on mop or allow to air dry.
- D. Use a spray device for spray application Spray 6-8 inches from the surface, rub with a brush, sponge or cloth. Avoid inhaling sprays.
- E. Rinse all surfaces that come in contact with food such as exterior of appliances, tables and stove top with potable water before rinse.

First aid procedure caused by chemical poisoning.

What to do?

- If the person has been exposed to poisonous fumes, such as carbon monoxide, get him or her into fresh air immediately.
- If the person swallowed the poison, remove anything remaining in the mouth.
- If the suspected poison is a household cleaner or other chemical, read the label and follow instructions for accident poisoning. If the product is toxic, the label will likely advise you to call the hospital/doctor.

- Follow treatment directions given by poison centers.

- If the poison spilled on the person's clothing, remove the clothing and pour the body with continuous tap water.

Ref: First Aid, Fourth Edition, American College of Emergency Physicians

Precaution: *Don't administer Ipecac syrup (a medicine that causes vomiting used to partially empty a person's stomach after a poison).*

The total facility cleaning and maintenance program of a food service department must be planned to reflect concern for sanitation as —a way of life. Facility sanitation results can be obtained through:

- ☐ Establishing high standards
- ☐ Rigid scheduling of assignments that are clearly understood by workers
- ☐ Ongoing training
- ☐ Proper use of cleaning supplies
- ☐ Provision of proper materials and equipment's to accomplish tasks, and
- ☐ Frequent meaningful inspections and performance reviews.

Regular cleaning of counter tops and floors needs to be done daily and is usually assigned as part of your regular daily duties. Other cleaning tasks that need to be done less frequently must be scheduled and assigned as needed for instance, daily, weekly, and monthly. General cleaning of floors, windows, walls and certain equipment should be assigned to personnel and it is often done in cooperation with the housekeeping and maintenance departments of the organizations

Each of the duties on the assignment list must be explained in detail on a written work sheet or —job breakdown‖ for the employee to follow. Job breakdown includes name of the task, tools and equipment and materials to be used, and a step-by-step list of what to do and how to do it. All food contact equipment ‘s, containers and utensils must be cleaned thoroughly after each use. This is especially true of meat grinders, slicers, cutting boards, knives, mixers, peelers, dishwashing machines and stationery can openers in order to prevent any cross-contamination.

How to Keep Your Kitchen Clean and Safe

Although they are not visible threats, the many micro-organisms waiting in your kitchen can infect your cooking and eating, and consequently have a negative effect on your health. Food poisoning and diarrhea are just some conditions which might be caused by preparing food in a dirty, germ-infested kitchen. In order to prevent these, you need to make sure that your kitchen is kept clean and safe from bacteria and other germs.

Here are ways to ensure the cleanliness of your kitchen and keep bacteria at bay.

1. Remove unnecessary clutter from surfaces. Discard appliances and gadgets that you don't use any more, as unused appliances would only tend to gather dust. Be sure that any appliances which are still maintained and used are stored in a safe place that is out of reach of children.

2. Keep your refrigerator clean and tidy. Before you do your weekly shopping, remove old and rotten food and clean shelves and racks. Look at expiration dates and be sure to label any containers of homemade food or leftovers. Don't make a habit of keeping food for too long in the fridge; rotten food will breed bacteria, and may even contaminate fresh food that is kept in the same area.

3. Use a rubbish bin with a lid to keep odors out that attracts flies and other insects. Empty it as soon as it smells, even if it is not yet full. You should ideally do this every day.

4. Wash and disinfect your rubbish bin once a week. This will ensure that any germs which might have remained even after you emptied out the rubbish will die and cannot multiply. Also, this will remove any foul smells which might be emanating from the trash bin.

5. Use separate chopping boards for different kinds of food. Keep separate chopping boards for your meat and your vegetables to reduce the spread of bacteria. Clean the meat chopping board extra carefully, especially after cutting chicken, as raw food has a higher tendency of containing bacteria.

6. Change the dishcloth you wipe surfaces with every day. Use a different cloth for surfaces than for dishes. Wash the cloth with hot water and, as appropriate, bleach. Use separate towels for hands and dishes, and change both of those regularly, too.

7. Use a paper towel for any mess that is particularly likely to cause contamination, such as raw meat or eggs, and anything that has fallen on the floor. It is better to use cleaning material that you can easily throw out after and prevent from contaminating other food, instead of using towels or sponges which you would still be likely to use again afterwards.

8. Keep kitchen floors free from debris and grease by sweeping and washing regularly. If something has spilled, make sure to mop it right away instead of letting the mess stay there. Aside from making sure that germs do not use the spillage as

a breeding ground, immediate mopping would also prevent unwanted accidents such as slipping from occurring.

9. Don't leave dirty crockery and pans to fester where they can attract harmful bacteria. Wash dishes with hot water and soap as soon as you're done using them.

10. Wash surfaces that get touched. Periodically wipe doorknobs, handles, buttons and controls, and light switches in and around your kitchen with cleaning agents. Even if they seem clean to the naked eye, they may already be harboring bacteria.

11. Wash your hands before handling food and again if you sneeze or cough, blow your nose, go to the bathroom, or touch high-use surfaces.

Safety measures

The following are safety measures that you should do when using cleaning agents:

1. Ensure adequate ventilation
2. Have knowledge of basic first aid
3. Wear cotton clothing to cover your limbs and other parts of your body that might be exposed to the cleaning agent.
4. Wear suitable footwear—it should be closed in and have a steel toe
5. Wear industrial strength, thick plastic or rubber gloves
6. Wear protective eye and face wear

Storage and security of chemicals

The following are recommendations for the storage and security of chemicals and cleaning agents:

1. Keep them in a separate area, away from food and other products
2. Keep on lower shelves to prevent accidents and to keep chemicals from falling into food products
3. Store in a cool, well-lit and well-ventilated room
4. Do not store near heat
5. Do not keep punctured aerosol cans
6. Store chemicals with lids tightly on
7. Make sure chemicals and other cleaning agents are clearly labelled, specifying their content and use

8. Ensure that the use by date or manufactured date is clearly readable
9. Storage containers should be free of corrosion and moisture
10. The storage area should be kept secure and locked when not in use
11. Always store chemicals in designated container
12. Do not mix chemicals